MultiOmicsSurvey Function Documentation

Auto-generated

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Contents

1 Introduction

This document provides comprehensive documentation for all custom functions used in the MultiOmicsSurvey project. Functions are organized by source file and presented in alphabetical order within each section.

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2 custom_functions.R

Functions for multi-omics data analysis, clustering evaluation, visualization, and data processing.

2.1 as.sunburstDF

```r
as.sunburstDF(DF, value_column = NULL, add_root = FALSE)
```

Description: Converts a hierarchical data frame into a sunburst-compatible format for visualization.

Parameters:

Parameter	Type	Description
`DF`	data.frame	Hierarchical data frame with category columns
`value_column`	character	Column name containing values for sizing
`add_root`	logical	Whether to add a root node (default: FALSE)

Returns: A data frame formatted for sunburst charts with `ids`, `labels`, `parents`, and `values` columns.

2.2 calculate_ari_index

```r
calculate_ari_index(cluster_df1, cluster_df2,
                    cluster_col1 = "Cluster", cluster_col2 = "Cluster",
                    sample_col1 = "samID", sample_col2 = "samID")
```

Description: Calculates the Adjusted Rand Index (ARI) between two clustering results.

Parameters:

Parameter	Type	Description
`cluster_df1`	data.frame	First clustering result
`cluster_df2`	data.frame	Second clustering result
`cluster_col1`	character	Column name for clusters in df1 (default: "Cluster")
`cluster_col2`	character	Column name for clusters in df2 (default: "Cluster")
`sample_col1`	character	Column name for sample IDs in df1 (default: "samID")
`sample_col2`	character	Column name for sample IDs in df2 (default: "samID")

Returns: Numeric value representing the Adjusted Rand Index (-1 to 1).

2.3 calculate_nmi_index

```r
calculate_nmi_index(cluster_df1, cluster_df2,
                    cluster_col1 = "Cluster", cluster_col2 = "Cluster",
                    sample_col1 = "samID", sample_col2 = "samID")
```

Description: Calculates the Normalized Mutual Information (NMI) between two clustering results.

Parameters:

Parameter	Type	Description
`cluster_df1`	data.frame	First clustering result
`cluster_df2`	data.frame	Second clustering result
`cluster_col1`	character	Column name for clusters in df1 (default: "Cluster")
`cluster_col2`	character	Column name for clusters in df2 (default: "Cluster")
`sample_col1`	character	Column name for sample IDs in df1 (default: "samID")
`sample_col2`	character	Column name for sample IDs in df2 (default: "samID")

Returns: Numeric value representing NMI (0 to 1).

2.4 calculate_S

```r
calculate_S(W)
```

Description: Calculates the S matrix from an affinity matrix W using the SNF transition probability formula.

Parameters:

Parameter	Type	Description
`W`	matrix	Affinity/similarity matrix

Returns: Normalized transition probability matrix S.

2.5 CIMLR_mod

```r
CIMLR_mod(X, c, no.dim = NA, k = 10, cores.ratio = 1, binary_flags = NULL,
          binary_distance = "binary", max_iter = 100, use_rsvd = FALSE)
```

Description: Modified CIMLR (Cancer Integration via Multi-kernel Learning) algorithm supporting mixed data types.

Parameters:

Parameter	Type	Description
`X`	list	List of data matrices (features $\times$ samples)
`c`	integer	Number of clusters
`no.dim`	integer	Number of dimensions for embedding (default: NA)
`k`	integer	Number of neighbors (default: 10)
`cores.ratio`	numeric	Ratio of cores for parallel processing (default: 1)
`binary_flags`	character	Vector indicating binary data types
`binary_distance`	character	Distance metric for binary data (default: "binary")
`max_iter`	integer	Maximum iterations (default: 100)
`use_rsvd`	logical	Use randomized SVD (default: FALSE)

Returns: List containing cluster assignments, similarity matrix, and convergence info.

2.6 CIMLR_Estimate_Number_of_Clusters_weight_mod

```
CIMLR_Estimate_Number_of_Clusters_weight_mod(all_data, NUMC = 2:5,
                                             cores.ratio = 0, weight,
                                             binary_flags = NULL, binary_distance = NULL)
```

Description: Estimates optimal number of clusters for weighted CIMLR using eigengap and rotation cost.

Parameters:

Parameter	Type	Description
`all_data`	list	List of data matrices
`NUMC`	integer	Range of cluster numbers to test (default: 2:5)
`cores.ratio`	numeric	Ratio of cores for parallel processing
`weight`	numeric	Feature weights
`binary_flags`	character	Vector indicating binary data types
`binary_distance`	character	Distance metric for binary data

Returns: List with optimal K estimates from eigengap and rotation methods.

2.7 CIMLR.weight_mod

```r
CIMLR.weight_mod(X, c, no.dim = NA, k = 10, cores.ratio = 0, weight,
                 binary_flags = NULL, binary_distance = NULL, max_iter = 100)
```

Description: Weighted CIMLR implementation with feature importance weights.

Parameters:

Parameter	Type	Description
X	list	List of data matrices
c	integer	Number of clusters
weight	numeric	Feature weights
(other parameters same as CIMLR_mod)		

Returns: Clustering result with weighted kernel integration.

2.8 coca_cc_mod

```r
coca_cc_mod(data = NULL, K = 2, B = 100, pItem = 0.8, clMethod = "hclust",
            innerLinkage = "complete", maxK = NULL)
```

Description: Modified COCA (Cluster of Clusters Analysis) consensus clustering.

Parameters:

Parameter	Type	Description
data	list	List of clustering matrices
K	integer	Number of final clusters (default: 2)
B	integer	Number of bootstrap iterations (default: 100)
pItem	numeric	Proportion of items to sample (default: 0.8)
clMethod	character	Clustering method (default: "hclust")
innerLinkage	character	Linkage for hierarchical clustering (default: "complete")
maxK	integer	Maximum K for consensus matrix

Returns: Consensus clustering result with cluster assignments.

2.9 compute_matrix_similarity

```r
compute_matrix_similarity(matrices)
```

Description: Computes pairwise similarity between multiple matrices using Frobenius norm and Pearson correlation.

Parameters:

Parameter	Type	Description
`matrices`	list	Named list of matrices to compare

Returns: List with Frobenius and Pearson similarity matrices.

2.10 compute_silhouette

```
compute_silhouette(cluster_df, similarity_matrix, normalize_matrix = FALSE)
```

Description: Computes silhouette scores from a similarity matrix and cluster assignments.

Parameters:

Parameter	Type	Description
`cluster_df`	data.frame	Data frame with sample IDs and cluster assignments
`similarity_matrix`	matrix	Sample similarity matrix
`normalize_matrix`	logical	Whether to normalize the matrix (default: FALSE)

Returns: Silhouette object with per-sample scores.

2.11 concordanceNetworkNMI_ANF

```
concordanceNetworkNMI_ANF(Wall, C, type)
```

Description: Calculates concordance between networks and clustering using NMI for ANF method.

Parameters:

Parameter	Type	Description
`Wall`	list	List of affinity matrices
`C`	integer	Number of clusters
`type`	character	Type of concordance calculation

Returns: NMI-based concordance scores.

2.12 create_annot_barchart

```
create_annot_barchart(plotdata = NULL, fill = NULL, title = NULL,
                      xlab = NULL, ylab = NULL, legend.position = "right")
```

Description: Creates annotated bar charts for visualizing cluster distributions.

Parameters:

Parameter	Type	Description
`plotdata`	data.frame	Data for plotting
`fill`	character	Column name for fill color
`title`	character	Plot title
`xlab`	character	X-axis label
`ylab`	character	Y-axis label
`legend.position`	character	Legend position (default: "right")

Returns: ggplot2 object.

2.13 create_density_curve

```
create_density_curve(matrix)
```

Description: Creates a density curve plot for matrix values.

Parameters:

Parameter	Type	Description
`matrix`	matrix	Input data matrix

Returns: ggplot2 density plot object.

2.14 create_density_plot_color

```
create_density_plot_color(matrix)
```

Description: Creates a colored density plot for matrix value distributions.

Parameters:

Parameter	Type	Description
`matrix`	matrix	Input data matrix

Returns: ggplot2 density plot with color gradient.

2.15 create_MO_heatmap

```
create_MO_heatmap(matrix = NULL, algorithm = NULL, clust_res = NULL,
                  data_name = NULL, fig.path = getwd(), ...)
```

Description: Creates multi-omics heatmaps with cluster annotations.

Parameters:

Parameter	Type	Description
`matrix`	matrix	Data or similarity matrix to plot
`algorithm`	character	Algorithm name for labeling
`clust_res`	data.frame	Clustering results with sample assignments
`data_name`	character	Name for the data type
`fig.path`	character	Output path for figures
`...`		Additional ComplexHeatmap parameters

Returns: ComplexHeatmap object saved to file.

2.16 dist2.cimlr_mod2

```
dist2.cimlr_mod2(x, c = NA, name = NULL, method = "sqeuclidean")
```

Description: Modified squared Euclidean distance calculation for CIMLR.

Parameters:

Parameter	Type	Description
`x`	matrix	Data matrix
`c`	matrix	Centers (optional)
`name`	character	Data name for labeling
`method`	character	Distance method (default: “sqeuclidean”)

Returns: Distance matrix.

2.17 dist2.cimlr.weight_mod

```
dist2.cimlr.weight_mod(x, weight, method = "sqeuclidean")
```

Description: Weighted distance calculation for CIMLR.

Parameters:

Parameter	Type	Description
`x`	matrix	Data matrix
`weight`	numeric	Feature weights
`method`	character	Distance method

Returns: Weighted distance matrix.

2.18 divide_by_quantile_normalize

```
divide_by_quantile_normalize(matrix, norm_quant = 0.05)
```

Description: Normalizes matrix by dividing by a quantile value.

Parameters:

Parameter	Type	Description
`matrix`	matrix	Input matrix
`norm_quant`	numeric	Quantile for normalization (default: 0.05)

Returns: Normalized matrix.

2.19 estimateNumberOfClustersGivenGraph_mod

```
estimateNumberOfClustersGivenGraph_mod(W, NUMC = 2:5)
```

Description: Estimates optimal cluster number using eigengap and rotation cost heuristics.

Parameters:

Parameter	Type	Description
`W`	matrix	Affinity/similarity matrix
`NUMC`	integer	Range of cluster numbers to evaluate (default: 2:5)

Returns: List with optimal K estimates.

2.20 fetch_citation

```
fetch_citation(algorithm)
```

Description: Retrieves citation information for a given algorithm.

Parameters:

Parameter	Type	Description
`algorithm`	character	Algorithm name

Returns: Character string with citation.

2.21 fetch_in_a_nutshell

```
fetch_in_a_nutshell(algorithm)
```

Description: Retrieves brief description for a given algorithm.

Parameters:

Parameter	Type	Description
`algorithm`	character	Algorithm name

Returns: Character string with algorithm summary.

2.22 frobenius_norm

```
frobenius_norm(mat1, mat2)
```

Description: Calculates the Frobenius norm distance between two matrices.

Parameters:

Parameter	Type	Description
`mat1`	matrix	First matrix
`mat2`	matrix	Second matrix

Returns: Numeric Frobenius norm value.

2.23 getMoHeatmap_prenorm_quart

```r
getMoHeatmap_prenorm_quart(data = NULL, is.binary = c(FALSE, FALSE, FALSE, FALSE, FALSE),
                           clust.res = NULL, ...)
```

Description: Generates multi-omics heatmap with quartile-based pre-normalization.

Parameters:

Parameter	Type	Description
`data`	list	List of omics data matrices
`is.binary`	logical	Vector indicating binary data types
`clust.res`	data.frame	Clustering results
`...`		Additional heatmap parameters

Returns: ComplexHeatmap object.

2.24 linear_quantile_normalize

```r
linear_quantile_normalize(matrix, norm_quant = 0.05)
```

Description: Linear quantile normalization scaling matrix to [0,1] based on quantiles.

Parameters:

Parameter	Type	Description
`matrix`	matrix	Input matrix
`norm_quant`	numeric	Quantile threshold (default: 0.05)

Returns: Normalized matrix.

2.25 mds_from_original_matrix

```r
mds_from_original_matrix(matrix = NULL, dist_method = NULL, algorithm = NULL,
                         clust_res = NULL, fig.path = getwd(), ...)
```

Description: Performs MDS from original data matrix for visualization.

Parameters:

Parameter	Type	Description
`matrix`	matrix	Input data matrix
`dist_method`	character	Distance method for MDS
`algorithm`	character	Algorithm name for labeling
`clust_res`	data.frame	Clustering results
`fig.path`	character	Output path

Returns: MDS plot saved to file.

2.26 MOVICS_jaccard_index

```
MOVICS_jaccard_index(clust1, clust2)
```

Description: Calculates Jaccard index between two cluster assignments.

Parameters:

Parameter	Type	Description
`clust1`	vector	First cluster assignment
`clust2`	vector	Second cluster assignment

Returns: Numeric Jaccard index (0 to 1).

2.27 multiple.kernel.cimlr_mod

```
multiple.kernel.cimlr_mod(x, cores.ratio = 1, name = NULL, method = "sqeuclidean")
```

Description: Generates multiple kernels for CIMLR integration.

Parameters:

Parameter	Type	Description
`x`	matrix	Data matrix
`cores.ratio`	numeric	Ratio of cores for parallel processing
`name`	character	Data name
`method`	character	Distance method

Returns: List of kernel matrices.

2.28 multiple.kernel.cimlr.weight_mod

```r
multiple.kernel.cimlr.weight_mod(x, cores.ratio = 0, weight, method = "sqeuclidean")
```

Description: Weighted multiple kernel generation for CIMLR.

Parameters:

Parameter	Type	Description
`x`	matrix	Data matrix
`weight`	numeric	Feature weights
`cores.ratio`	numeric	Ratio of cores
`method`	character	Distance method

Returns: Weighted kernel list.

2.29 nemo.affinity.graph_mod

```r
nemo.affinity.graph_mod(raw.data, k = NA, sigma = 0.5,
                        binary_flags = rep("No", length(raw.data)),
                        binary_distance = NULL)
```

Description: Modified NEMO affinity graph construction supporting mixed data types.

Parameters:

Parameter	Type	Description
`raw.data`	list	List of data matrices
`k`	integer	Number of neighbors
`sigma`	numeric	Kernel width (default: 0.5)
`binary_flags`	character	Vector indicating binary data
`binary_distance`	character	Distance for binary data

Returns: Affinity matrix.

2.30 normalize_affinity_matrix

```r
normalize_affinity_matrix(W)
```

Description: Normalizes an affinity matrix to have row sums of 1.

Parameters:

Parameter	Type	Description
W	matrix	Affinity matrix

Returns: Row-normalized affinity matrix.

2.31 pca_from_original_matrix

```
pca_from_original_matrix(mydata = NULL, algorithm = NULL, clust_res = NULL,
                         data_name = NULL, fig.path = getwd(), ...)
```

Description: Performs PCA on original data and creates visualization with cluster colors.

Parameters:

Parameter	Type	Description
mydata	matrix	Input data matrix
algorithm	character	Algorithm name for labeling
clust_res	data.frame	Clustering results
data_name	character	Data type name
fig.path	character	Output path

Returns: PCA plot saved to file.

2.32 pca_from_sim_matrix

```
pca_from_sim_matrix(sim_matrix = NULL, algorithm = NULL, clust_res = NULL,
                    data_name = NULL, fig.path = getwd(), ...)
```

Description: Performs PCA-based visualization from a similarity matrix.

Parameters:

Parameter	Type	Description
sim_matrix	matrix	Similarity/affinity matrix
algorithm	character	Algorithm name
clust_res	data.frame	Clustering results
data_name	character	Data type name
fig.path	character	Output path

Returns: PCA plot from similarity matrix.

2.33 pearson_correlation

```
pearson_correlation(mat1, mat2)
```

Description: Calculates Pearson correlation between vectorized matrices.

Parameters:

Parameter	Type	Description
`mat1`	matrix	First matrix
`mat2`	matrix	Second matrix

Returns: Pearson correlation coefficient.

2.34 plot_MOFA_cumul_var_exp

```
plot_MOFA_cumul_var_exp(object, factor_range = NULL, ...)
```

Description: Plots cumulative variance explained by MOFA factors.

Parameters:

Parameter	Type	Description
`object`	MOFA object	Trained MOFA model
`factor_range`	integer	Range of factors to plot
`...`		Additional ggplot parameters

Returns: ggplot2 cumulative variance plot.

2.35 plot_MOFA_var_exp

```
plot_MOFA_var_exp(object, factor_range = NULL, ...)
```

Description: Plots variance explained per view and factor for MOFA models.

Parameters:

Parameter	Type	Description
`object`	MOFA object	Trained MOFA model
`factor_range`	integer	Range of factors to plot
`...`		Additional ggplot parameters

Returns: ggplot2 variance explained heatmap.

2.36 rankFeaturesByNMI_parallely

```
rankFeaturesByNMI_parallely(data, W, ncores = detectCores() - 1,
                            binary = FALSE, nn = NULL, sigma)
```

Description: Ranks features by NMI contribution using parallel processing.

Parameters:

Parameter	Type	Description
`data`	list	Single-element list with data matrix
`W`	matrix	Similarity matrix
`ncores`	integer	Number of cores (default: detectCores() - 1)
`binary`	logical	Binary data flag
`nn`	integer	Number of neighbors
`sigma`	numeric	Kernel width

Returns: Data frame with feature NMI scores.

2.37 rankFeaturesByNMI_parallely_ANF

```
rankFeaturesByNMI_parallely_ANF(data, W, ncores = detectCores() - 1,
                                binary = FALSE, ...)
```

Description: ANF-specific feature ranking by NMI.

Parameters:

Parameter	Type	Description
`data`	list	Data list
`W`	matrix	ANF similarity matrix
`ncores`	integer	Number of cores
`binary`	logical	Binary data flag

Returns: Feature ranking data frame.

2.38 reshape_Pearson_for_tests

```
reshape_Pearson_for_tests(similarities)
```

Description: Reshapes Pearson correlation matrix for statistical testing.

Parameters:

Parameter	Type	Description
`similarities`	matrix	Similarity matrix

Returns: Reshaped data frame for testing.

2.39 standardize_rows

```r
standardize_rows(mat)
```

Description: Z-score standardizes each row of a matrix.

Parameters:

Parameter	Type	Description
`mat`	matrix	Input matrix

Returns: Row-standardized matrix.

2.40 TCGAanalyze_survival_custom3

```r
TCGAanalyze_survival_custom3(clinical_data, cluster_col, time_col,
                             event_col, algorithm = NULL, ...)
```

Description: Custom survival analysis for TCGA data with multi-group comparisons.

Parameters:

Parameter	Type	Description
`clinical_data`	data.frame	Clinical data with survival information
`cluster_col`	character	Column with cluster assignments
`time_col`	character	Column with survival time
`event_col`	character	Column with event status
`algorithm`	character	Algorithm name for labeling

Returns: Survival analysis results with Kaplan-Meier plots.

---

3 fast_pathfindR_hclust.R

Optimized functions for pathway enrichment term clustering.

3.1 cluster_enriched_terms_fast

```r
cluster_enriched_terms_fast(enrichment_res, method = "hierarchical",
                            plot_clusters_graph = TRUE, ...)
```

Description: Main function for clustering enriched pathway terms using optimized algorithms.

Parameters:

Parameter	Type	Description
`enrichment_res`	data.frame	Enrichment results from pathfindR
`method`	character	Clustering method: "hierarchical" or "fuzzy"
`plot_clusters_graph`	logical	Whether to plot cluster graph
`...`		Additional clustering parameters

Returns: List with clustered terms and visualization.

3.2 cluster_graph_vis_fast

```r
cluster_graph_vis_fast(clu_obj, kappa_mat, enrichment_res,
                       kappa_threshold = 0.35, ...)
```

Description: Creates fast graph visualization of term clusters.

Parameters:

Parameter	Type	Description
`clu_obj`	list	Clustering result object
`kappa_mat`	matrix	Kappa statistic matrix
`enrichment_res`	data.frame	Original enrichment results
`kappa_threshold`	numeric	Threshold for edge creation

Returns: igraph visualization object.

3.3 create_kappa_matrix_fast

```r
create_kappa_matrix_fast(enrichment_res, use_description = FALSE,
                         use_active_snw_genes = FALSE)
```

Description: Creates kappa statistic matrix for pathway term similarity using optimized algorithm.

Parameters:

Parameter	Type	Description
`enrichment_res`	data.frame	Enrichment results
`use_description`	logical	Use pathway descriptions
`use_active_snw_genes`	logical	Use active subnetwork genes

Returns: Kappa matrix for term similarity.

3.4 fuzzy_term_clustering_fast

```r
fuzzy_term_clustering_fast(kappa_mat, enrichment_res,
                           kappa_threshold = 0.35, ...)
```

Description: Performs fuzzy clustering on pathway terms.

Parameters:

Parameter	Type	Description
`kappa_mat`	matrix	Kappa similarity matrix
`enrichment_res`	data.frame	Enrichment results
`kappa_threshold`	numeric	Similarity threshold

Returns: Fuzzy clustering assignments.

3.5 hierarchical_term_clustering_fast

```r
hierarchical_term_clustering_fast(kappa_mat, enrichment_res,
                                  num_clusters = NULL, ...)
```

Description: Performs hierarchical clustering on pathway terms.

Parameters:

Parameter	Type	Description
`kappa_mat`	matrix	Kappa similarity matrix

Parameter	Type	Description
`enrichment_res`	data.frame	Enrichment results
`num_clusters`	integer	Number of clusters (auto if NULL)

Returns: Hierarchical clustering result.

---

4 modified_MOVICS_functions.R

Modified MOVICS package functions for multi-omics analysis.

4.1 compAgree2

```
compAgree2(moic.res = NULL, subt2comp = NULL, doPlot = TRUE, ...)
```

Description: Compares agreement between two clustering results with visualization.

Parameters:

Parameter	Type	Description
`moic.res`	list	MOVICS result object
`subt2comp`	data.frame	Second clustering to compare
`doPlot`	logical	Generate comparison plot

Returns: Agreement statistics and optional Sankey plot.

4.2 compAgree_single_algorithm

```
compAgree_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                           subt2comp = NULL, ...)
```

Description: Single-algorithm version of clustering agreement comparison.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`moic.res`	list	Clustering result
`subt2comp`	data.frame	Reference clustering

Returns: Agreement metrics.

4.3 compClinvar2

```
compClinvar2(moic.res = NULL, var2comp = NULL, strata = NULL,
             factorVars = NULL, ...)
```

Description: Compares clinical variables across subtypes.

Parameters:

Parameter	Type	Description
`moic.res`	list	MOVICS result
`var2comp`	data.frame	Clinical variables
`strata`	character	Stratification variable
`factorVars`	character	Factor variables

Returns: Clinical comparison table with p-values.

4.4 compClinvar_ordinal_single_algorithm

```
compClinvar_ordinal_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                                     var2comp = NULL, ordinal_vars = NULL, ...)
```

Description: Compares ordinal clinical variables using appropriate tests.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`moic.res`	list	Clustering result
`var2comp`	data.frame	Clinical variables
`ordinal_vars`	character	Ordinal variable names

Returns: Ordinal comparison results.

4.5 compClinvar_single_algorithm

```
compClinvar_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                             var2comp = NULL, ...)
```

Description: Single-algorithm clinical variable comparison.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`moic.res`	list	Clustering result
`var2comp`	data.frame	Clinical variables

Returns: Comparison table.

4.6 compDrugsen_single_algorithm

```
compDrugsen_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                             norm.expr = NULL, drugs = c("Cisplatin", "Paclitaxel"), ...)
```

Description: Compares predicted drug sensitivity across subtypes.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`moic.res`	list	Clustering result
`norm.expr`	matrix	Normalized expression
`drugs`	character	Drug names to analyze

Returns: Drug sensitivity comparison.

4.7 compFGA_mod

```
compFGA_mod(moic.res = NULL, segment = NULL, iscopynumber = FALSE,
            ga_column = NULL, ...)
```

Description: Compares Fraction of Genome Altered (FGA) across subtypes.

Parameters:

Parameter	Type	Description
`moic.res`	list	Clustering result
`segment`	data.frame	Segmentation data
`iscopynumber`	logical	Whether data is copy number
`ga_column`	character	Genome alteration column

Returns: FGA comparison with visualization.

4.8 compFGA_optimized

```r
compFGA_optimized(moic.res = NULL, segment = NULL, iscopynumber = FALSE,
                  ga_column = NULL, ...)
```

Description: Optimized FGA comparison with improved performance.

Parameters:

Parameter	Type	Description
(Same as compFGA_mod)		

Returns: FGA comparison results.

4.9 compMut_single_algorithm

```r
compMut_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                         mut.matrix = NULL, freq.cutoff = 0.05, ...)
```

Description: Compares mutation frequencies across subtypes.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`moic.res`	list	Clustering result
`mut.matrix`	matrix	Mutation matrix
`freq.cutoff`	numeric	Frequency cutoff (default: 0.05)

Returns: Mutation comparison with oncoprint.

4.10 compMut_single_algorithm_sc

```r
compMut_single_algorithm_sc(algorithm_name = "CS", moic.res = NULL,
                            mut.matrix = NULL, ...)
```

Description: Single-cell adapted mutation comparison.

Parameters:

Parameter	Type	Description
(Same as compMut_single_algorithm)		

Returns: Mutation comparison for single-cell data.

4.11 compSurv_ext

```
compSurv_ext(moic.res = NULL, surv.info = NULL, convt.time = "d", ...)
```

Description: Extended survival comparison with multiple tests.

Parameters:

Parameter	Type	Description
`moic.res`	list	Clustering result
`surv.info`	data.frame	Survival information
`convt.time`	character	Time conversion ("d", "m", "y")

Returns: Comprehensive survival analysis.

4.12 getMoHeatmap_single_algorithm

```
getMoHeatmap_single_algorithm(algorithm_name = "CS", data = NULL,
                              is.binary = c(FALSE), ...)
```

Description: Generates multi-omics heatmap for single algorithm result.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`data`	list	Multi-omics data
`is.binary`	logical	Binary data flags

Returns: ComplexHeatmap object.

4.13 getMoHeatmap_single_algorithm2

```r
getMoHeatmap_single_algorithm2(algorithm_name = "CS", data = NULL, ...)
```

Description: Alternative multi-omics heatmap generation.

Parameters:

Parameter	Type	Description
(Same as getMoHeatmap_single_algorithm)		

Returns: Heatmap visualization.

4.14 getSilhouette_ggplot

```r
getSilhouette_ggplot(sil = NULL, clust.col = NULL, fig.path = getwd(),
                     fig.name = "silhouette_ggplot", algorithm = "", ...)
```

Description: Creates ggplot2-based silhouette plot.

Parameters:

Parameter	Type	Description
`sil`	silhouette	Silhouette object
`clust.col`	character	Cluster colors
`fig.path`	character	Output path
`fig.name`	character	Output filename
`algorithm`	character	Algorithm name for labels

Returns: ggplot2 silhouette visualization.

4.15 ntp_mod

```r
ntp_mod(emat, templates, nPerm = 1000, distance = "cosine", ...)
```

Description: Modified Nearest Template Prediction algorithm.

Parameters:

Parameter	Type	Description
`emat`	matrix	Expression matrix
`templates`	list	Template signatures
`nPerm`	integer	Number of permutations

Parameter	Type	Description
`distance`	character	Distance metric

Returns: NTP predictions with p-values.

4.16 plot_pathway_heatmaps

```
plot_pathway_heatmaps(gsea.lists, norm.expr = NULL, algorithm_name = NULL, ...)
```

Description: Creates pathway-level heatmaps from GSEA results.

Parameters:

Parameter	Type	Description
`gsea.lists`	list	GSEA result lists
`norm.expr`	matrix	Normalized expression
`algorithm_name`	character	Algorithm identifier

Returns: Pathway heatmap visualizations.

4.17 prepare_gsea_output_for_hclust

```
prepare_gsea_output_for_hclust(gsea_output, dgea_output_name_style = "", ...)
```

Description: Prepares GSEA output for hierarchical clustering analysis.

Parameters:

Parameter	Type	Description
`gsea_output`	list	GSEA results
`dgea_output_name_style`	character	DGEA naming convention

Returns: Formatted data for pathway clustering.

4.18 runDEA_mod

```r
runDEA_mod(dea.method = c("deseq2", "edger", "limma"), expr = NULL,
           moic.res = NULL, ...)
```

Description: Modified differential expression analysis wrapper.

Parameters:

Parameter	Type	Description
`dea.method`	character	DEA method to use
`expr`	matrix	Expression matrix
`moic.res`	list	Clustering result

Returns: DEA results for all pairwise comparisons.

4.19 runGSEA_mod

```r
runGSEA_mod(moic.res = NULL, dea.method = c("deseq2", "edger", "limma"), ...)
```

Description: Modified GSEA analysis for MOVICS results.

Parameters:

Parameter	Type	Description
`moic.res`	list	Clustering result
`dea.method`	character	DEA method used

Returns: GSEA enrichment results.

4.20 runGSEA_mod_4.4

```r
runGSEA_mod_4.4(moic.res = NULL, dea.method = c("deseq2", "edger", "limma"), ...)
```

Description: GSEA modified for R 4.4 compatibility.

Parameters:

Parameter	Type	Description
(Same as runGSEA_mod)		

Returns: R 4.4 compatible GSEA results.

4.21 runGSEA_mod_4.4_single_algorithm

```
runGSEA_mod_4.4_single_algorithm(algorithm_name = "CS", moic.res = NULL, ...)
```

Description: Single-algorithm GSEA for R 4.4.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`moic.res`	list	Clustering result

Returns: GSEA results with algorithm prefix.

4.22 runGSVA_mod_4.4

```
runGSVA_mod_4.4(moic.res = NULL, norm.expr = NULL, gset.gmt.path = NULL, ...)
```

Description: Modified GSVA analysis for R 4.4.

Parameters:

Parameter	Type	Description
`moic.res`	list	Clustering result
`norm.expr`	matrix	Normalized expression
`gset.gmt.path`	character	Path to GMT file

Returns: GSVA enrichment scores.

4.23 runGSVA_mod_4.4_single_algorithm

```
runGSVA_mod_4.4_single_algorithm(algorithm_name = "CS", moic.res = NULL, ...)
```

Description: Single-algorithm GSVA for R 4.4.

Parameters:

Parameter	Type	Description
(Same as runGSVA_mod_4.4 with algorithm_name)		

Returns: GSVA results with algorithm prefix.

4.24 runKappa_single_algorithm

```
runKappa_single_algorithm(algorithm_name = "CS", moic.res = NULL, ...)
```

Description: Calculates Cohen's Kappa for clustering agreement.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`moic.res`	list	Clustering result

Returns: Kappa statistics.

4.25 runMarker_mod_4.4

```
runMarker_mod_4.4(moic.res = NULL, dea.method = c("deseq2", "edger", "limma"), ...)
```

Description: Marker gene identification for R 4.4.

Parameters:

Parameter	Type	Description
`moic.res`	list	Clustering result
`dea.method`	character	DEA method

Returns: Marker genes per subtype.

4.26 runMarker_single_algorithm

```
runMarker_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                           dea.method = c("deseq2", "edger", "limma"), ...)
```

Description: Single-algorithm marker identification.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier
`moic.res`	list	Clustering result
`dea.method`	character	DEA method

Returns: Marker genes with algorithm prefix.

4.27 runMarker_single_algorithm_no_export

```
runMarker_single_algorithm_no_export(algorithm_name = "CS", moic.res = NULL, ...)
```

Description: Marker identification without file export.

Parameters:

Parameter	Type	Description
(Same as runMarker_single_algorithm)		

Returns: In-memory marker results.

4.28 runNTP_mod

```
runNTP_mod(expr = NULL, templates = NULL, scaleFlag = TRUE,
           centerFlag = TRUE, ...)
```

Description: Modified Nearest Template Prediction.

Parameters:

Parameter	Type	Description
`expr`	matrix	Expression matrix
`templates`	list	Template gene sets
`scaleFlag`	logical	Scale data
`centerFlag`	logical	Center data

Returns: NTP classification results.

4.29 runPAM_single_algorithm

```
runPAM_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                        norm.expr = NULL, ...)
```

Description: PAM (Prediction Analysis for Microarrays) for subtype prediction.

Parameters:

Parameter	Type	Description
`algorithm_name`	character	Algorithm identifier

Parameter	Type	Description
`moic.res`	list	Clustering result
`norm.expr`	matrix	Normalized expression

Returns: PAM classifier and predictions.

4.30 subHeatmap_mod

```
subHeatmap_mod(emat, res, templates, keepN = TRUE, labRow = NULL, ...)
```

Description: Modified subtype heatmap visualization.

Parameters:

Parameter	Type	Description
`emat`	matrix	Expression matrix
`res`	list	NTP results
`templates`	list	Template signatures
`keepN`	logical	Keep all samples
`labRow`	character	Row labels

Returns: Subtype heatmap.

4.31 twoclassdeseq2_mod

```
twoclassdeseq2_mod(moic.res = NULL, countsTable = NULL, prefix = NULL, ...)
```

Description: Two-class DESeq2 differential expression.

Parameters:

Parameter	Type	Description
`moic.res`	list	Clustering result
`countsTable`	matrix	Count matrix
`prefix`	character	Output prefix

Returns: DESeq2 results.

4.32 twoclassedger_mod

```r
twoclassedger_mod(moic.res = NULL, countsTable = NULL, prefix = NULL, ...)
```

Description: Two-class edgeR differential expression.

Parameters:

Parameter	Type	Description
(Same as twoclassdeseq2_mod)		

Returns: edgeR results.

4.33 twoclasslimma_mod

```r
twoclasslimma_mod(moic.res = NULL, norm.expr = NULL, prefix = NULL, ...)
```

Description: Two-class limma differential expression.

Parameters:

Parameter	Type	Description
`moic.res`	list	Clustering result
`norm.expr`	matrix	Normalized expression
`prefix`	character	Output prefix

Returns: limma results.

---

5 Spectrum_functions.R

Modified Spectrum spectral clustering functions.

5.1 CNN_kernel_mod

```r
CNN_kernel_mod(mat, NN = 3, NN2 = 7, distance = "euclidean")
```

Description: Modified Continuous Nearest Neighbor kernel construction.

Parameters:

Parameter	Type	Description
`mat`	matrix	Data matrix
`NN`	integer	Number of nearest neighbors (default: 3)
`NN2`	integer	Extended neighbors (default: 7)
`distance`	character	Distance metric

Returns: CNN kernel matrix.

5.2 kernfinder_local_mod

```
kernfinder_local_mod(data, maxk = 10, fontsize = 12, silent = FALSE, ...)
```

Description: Local kernel parameter finder for Spectrum.

Parameters:

Parameter	Type	Description
`data`	matrix	Data matrix
`maxk`	integer	Maximum k to test (default: 10)
`fontsize`	integer	Plot font size
`silent`	logical	Suppress output

Returns: Optimal local kernel parameters.

5.3 kernfinder_mine_mod

```
kernfinder_mine_mod(data, maxk = 10, fontsize = 12, silent = FALSE, ...)
```

Description: Mining-based kernel parameter finder.

Parameters:

Parameter	Type	Description
(Same as kernfinder_local_mod)		

Returns: Optimal kernel parameters.

5.4 rbfkernel_b_mod

```r
rbfkernel_b_mod(mat, K = 3, sigma = 1, distance = "euclidean")
```

Description: Modified RBF kernel construction.

Parameters:

Parameter	Type	Description
`mat`	matrix	Data matrix
`K`	integer	Number of neighbors (default: 3)
`sigma`	numeric	Kernel width (default: 1)
`distance`	character	Distance metric

Returns: RBF kernel matrix.

5.5 Spectrum_bin_and_par

```r
Spectrum_bin_and_par(data, method = "CNN", maxk = 10, ...)
```

Description: Main Spectrum clustering with binary data support and parallel processing.

Parameters:

Parameter	Type	Description
`data`	list	List of data matrices
`method`	character	Kernel method ("CNN", "RBF")
`maxk`	integer	Maximum clusters to test
`...`		Additional Spectrum parameters

Returns: Spectrum clustering results with cluster assignments and metrics.

---

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